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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 6

Title: Creating HR Analytics Dashboard

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Experiment No.: 6

Creating HR Attrition Analytics Dashboard

Problem Statement

You are an HR Data Analyst in a multinational organization. The HR department wants insights into employee demographics, hiring trends, attrition, and workforce distribution. Develop an interactive HR Analytics Dashboard in Power BI to support workforce planning and employee management using the IBM HR Analytics Employee Attrition dataset.

Methodology / Steps Followed

Step 1: Download and Collect the Dataset

First, I downloaded the IBM HR Analytics Employee Attrition Dataset from Kaggle. The dataset was saved in CSV format and imported into Power BI for analysis.

The dataset contains employee-related information such as Age, Gender, Department, Job Role, Monthly Income, Business Travel, Education, Job Level, Job Satisfaction, Overtime, Years at Company, and Attrition.

The dataset contains 1,470 employee records.

Screenshot 1:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education
1	41	Yes	Travel_Rarely	1102	Sales	1	
2	49	No	Travel_Frequently	279	Research & Development	8	
3	37	Yes	Travel_Rarely	1373	Research & Development	2	
4	33	No	Travel_Frequently	1392	Research & Development	3	
5	27	No	Travel_Rarely	591	Research & Development	2	
6	32	No	Travel_Frequently	1005	Research & Development	2	
7	59	No	Travel_Rarely	1324	Research & Development	3	
8	30	No	Travel_Rarely	1358	Research & Development	24	
9	38	No	Travel_Frequently	216	Research & Development	23	
10	36	No	Travel_Rarely	1299	Research & Development	27	
11	35	No	Travel_Rarely	809	Research & Development	16	
12	29	No	Travel_Rarely	153	Research & Development	15	
13	31	No	Travel_Rarely	670	Research & Development	26	
14	34	No	Travel_Rarely	1346	Research & Development	19	
15	28	Yes	Travel_Rarely	101	Research & Development	24	
16	29	No	Travel_Rarely	1389	Research & Development	21	
17	32	No	Travel_Rarely	334	Research & Development	5	
18	22	No	Non-Travel	1121	Research & Development	16	
19	53	No	Travel_Rarely	2219	Sales	2	
20	38	No	Travel_Rarely	371	Research & Development	2	
21	24	No	Non-Travel	673	Research & Development	11	
22	36	Yes	Travel_Rarely	1218	Sales	9	
23	34	No	Travel_Rarely	419	Research & Development	7	
24	21	No	Travel_Rarely	391	Research & Development	15	
25	34	No	Travel_Rarely	698	Research & Development	7	

Step 2: Data Preparation

After importing the dataset into Power BI, I checked the data and verified the data types of different columns.

The following types of columns were checked:

- Age → Whole Number
- Attrition → Text
- BusinessTravel → Text
- Department → Text
- DistanceFromHome → Whole Number
- Education → Whole Number
- EmployeeNumber → Whole Number
- JobLevel → Whole Number
- MonthlyIncome → Whole Number
- OverTime → Text
- PerformanceRating → Whole Number
- YearsAtCompany → Whole Number

I also checked the dataset for missing values and data errors. The data was prepared so that it could be used for creating HR analytics visualizations.

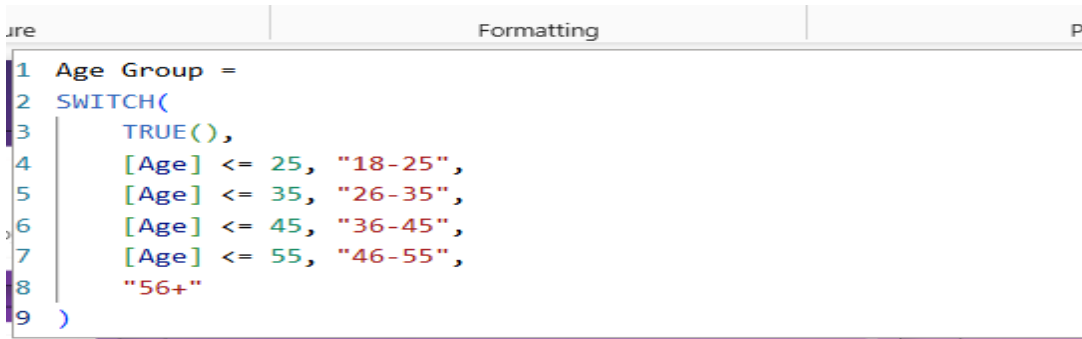
Step 3: Create Calculated Column

To perform better analysis of employee attrition based on age, I created an **Age Group** column using DAX.

The employees were divided into different age groups such as:

The calculated column was created using the following DAX formula:

Screenshot 3:



```
1 Age Group =
2 SWITCH(
3     TRUE(),
4     [Age] <= 25, "18-25",
5     [Age] <= 35, "26-35",
6     [Age] <= 45, "36-45",
7     [Age] <= 55, "46-55",
8     "56+"
9 )
```

Step 4: Create DAX Measures

To display important HR metrics, I created the following DAX measures.

Total Employees

Total Employees = COUNTROWS('HR-employee_Attrition-assignment')

This measure calculates the total number of employees in the dataset.

Employees Left

Employees Left = CALCULATE(COUNTROWS('HR-employee_Attrition-assignment'),
'HR-employee_Attrition-assignment'[Attrition] = "Yes")

This measure calculates the number of employees who left the organization.

Attrition Rate

Attrition Rate = DIVIDE([Employees Left], [Total Employees], 0)

This measure calculates the percentage of employees who left the organization.

Average Age

Average Age = AVERAGE('HR-employee_Attrition-assignment'[Age])

This measure calculates the average age of employees.

Average Monthly Income

Average_Monthly_Incom= AVERAGE('HR-employee_Atterition-assignment'[MonthlyIncome])

This measure calculates the average monthly income of employees.

These measures were used to create KPI cards on the dashboard.

Screenshot 4:

```
1 Attrition Rate =  
2 DIVIDE(  
3 | [Employees Left],  
4 | [Total Employees])
```

```
1 Average Age =  
2 AVERAGE('HR-employee_Atterition-assignment6'[Age])
```

```
1 Average Monthly Income =  
2 AVERAGE('HR-employee_Atterition-assignment6'[MonthlyIncome])
```

```
1 Employees Left =  
2 CALCULATE(  
3 | COUNTROWS('HR-employee_Atterition-assignment6'),  
4 | 'HR-employee_Atterition-assignment6'[Attrition] = "Yes"  
5 )
```

```
1 Total Employees = COUNTROWS('HR-employee_Atterition-assignment6')
```

Step 5: Dashboard Development

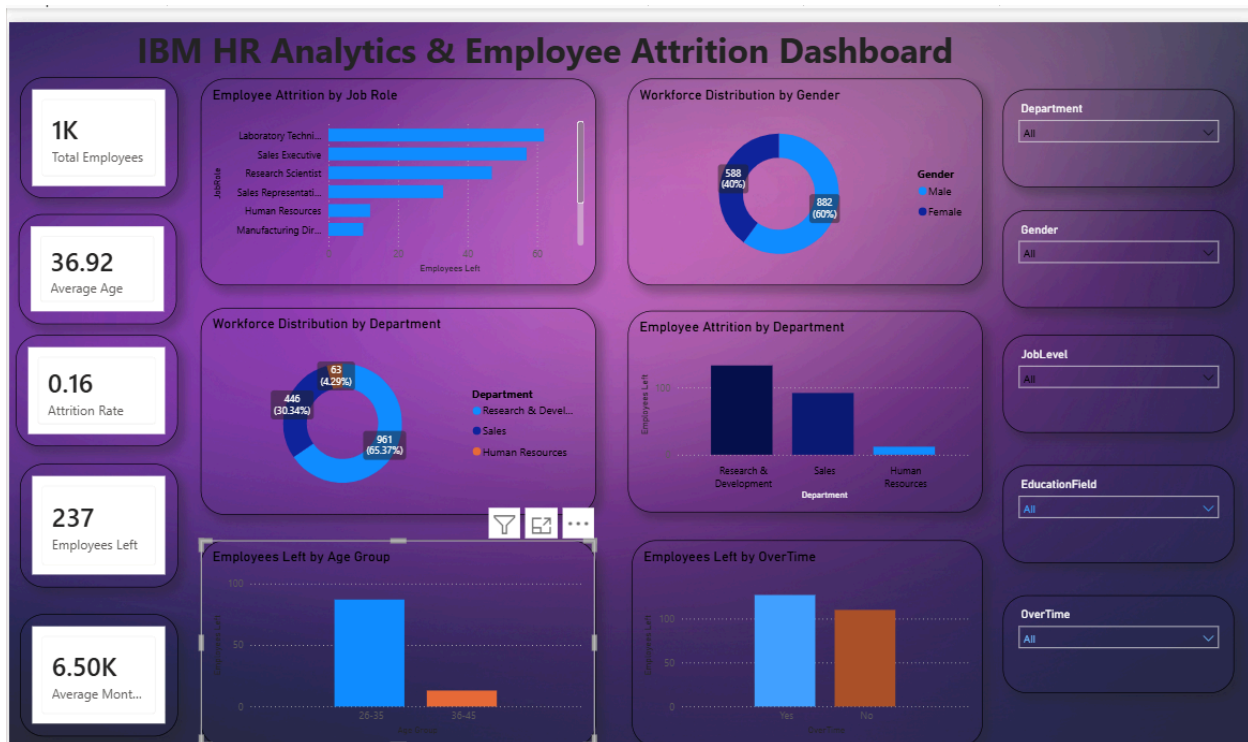
After preparing the dataset and creating DAX measures, I designed an interactive HR Analytics Dashboard in Power BI.

The dashboard contains the following visualizations:

- KPI Card showing Total Employees.
- KPI Card showing Average Age.
- KPI Card showing Attrition Rate.
- KPI Card showing Employees Left.
- KPI Card showing Average Monthly Income.
- Bar Chart showing Employee Attrition by Job Role.
- Donut Chart showing Workforce Distribution by Gender.
- Donut Chart showing Workforce Distribution by Department.
- Column Chart showing Employee Attrition by Department.
- Column Chart showing Employee Attrition by Age Group.
- Column Chart showing Employee Attrition by Overtime.
- Interactive Slicers for Department, Gender, Job Level, Education Field, and Overtime.

These visualizations provide an overall view of employee demographics and attrition patterns.

Screenshot 5:



Business Insights

From the dashboard, the following observations were made:

- The organization has 1,470 employees in the dataset.
- A total of 237 employees have left the organization.
- The overall attrition rate is approximately 16.12%.
- The average age of employees is approximately 36.92 years.
- The average monthly income is approximately 6,500.
- Research & Development has the largest workforce among the departments.
- Research & Development also has the highest number of employees who left.
- The organization has 882 male employees and 588 female employees.
- Employee attrition varies across different job roles.
- Employees working overtime show a relatively high level of attrition.
- Interactive slicers allow HR managers to analyze specific employee groups and departments.

Conclusion

In this assignment, an interactive HR Analytics Dashboard was successfully developed using Power BI and the IBM HR Analytics Employee Attrition dataset.

The dataset was imported and prepared before creating calculated columns and DAX measures. Different visualizations such as KPI cards, bar charts, column charts, donut charts, and interactive slicers were used to analyze employee demographics, workforce distribution, and employee attrition.

The dashboard provides meaningful HR insights and makes it easier to identify departments, job roles, age groups, and working conditions associated with employee attrition. It can support HR managers in making better decisions related to employee retention and workforce planning.

Power BI GITHUB Link

<https://github.com/Amina872/Power-BI/blob/main/IBM-HR-Employee-Attrition-Dashboard-assignment-6.pbix>

